

Transfer of Training Trends and Future Research Direction: Insight from Bibliometric Analysis

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Abstract: *The transfer of training remains a critical concern in organizational learning and development. Despite decades of empirical research, the evolving complexity of training contexts has prompted renewed scholarly attention. This study presents a comprehensive bibliometric analysis of global research on transfer of training published between 1909–2025. Using Scopus-indexed data and employing VOSviewer and Publish or Perish software, this review maps the intellectual structure, identifies influential authors, journals, and institutions, and uncovers thematic clusters and emerging trends. The study spans 116 years and collected 2754 documents. The findings indicated that there are six clusters associated with transfer of training at the workplace: 1) determinants of training transfer, 2) AI and immersive technologies for training and skill transfer, 3) cognitive function and training interventions in learning transfer, 4) cognitive function and training interventions in learning transfer, 5) applied learning and simulation in training transfer, and 6) meta-analytic perspectives on motivation and transfer. In addition, several future research questions are proposed for further research directions for scholars.*

Keywords: Transfer of Training; Training Transfer; Future Research; Bibliometrics; Network Analysis

1. Introduction

Today's workplaces are experiencing unprecedented changes driven by automation, remote work, digital collaboration tools, and continuous learning demands (Žur & Friedl, 2021). Consequently, organizations are shifting from traditional training models toward more agile, technology-enhanced learning solutions such as virtual simulations, mobile learning, AI-based adaptive learning, and microlearning platforms (Kambur & Yildirim, 2022; Shukla et al., 2023). While these innovations can improve learning accessibility and personalisation, their efficacy in encouraging transfer is dependent on how well they match job objectives and organisational support mechanisms (Capinera, 2021). Despite this digital transition, the core difficulty remains the same, converting what employees learn during training into increased job performance (Quesada-Pallarès et al., 2022). Organisations are increasingly emphasising "learning agility," worker reskilling, and performance-based learning outcomes, elevating training transfer to a strategic goal.

Transfer of training is the application, maintenance, and generalisation of knowledge, abilities, and attitudes acquired in a training context to the actual work environment (Fatima & Siddiqui, 2020). In other words, it refers to how well and consistently what was learnt during training is applied on the job. Individual traits, training design, and the work environment all have an impact on training transfer (Suhepi & Syah, 2018). Organisations invest heavily in employee training with the hope that newly learnt information, skills, and behaviours would be implemented in the workplace to increase performance and productivity (Al-Swidi & Al Yahya, 2017). However, research has shown that the extent of transfer is often limited, with a significant proportion of training content failing to be applied in practice. This gap between learning and workplace application is a long-standing concern in human resource development and organisational psychology. Understanding the factors that facilitate, or hinder transfer is therefore essential for designing effective training interventions. Moreover, improving transfer of training has direct implications for maximising return on investment in training programmes and sustaining competitive advantage (Mohamad Yunus et al., 2022).

Despite considerable training expenditures across industries, empirical studies regularly show a transfer gap, with only a minority of learnt competencies being implemented in the workplace (Blume et al., 2019). Empirical evidence consistently shows a disparity between training delivery and training transfer, with estimates indicating that only a small percentage often less than 40% of the learning material is effectively transferred to the workplace (Chee et al., 2018; Kodwani, 2017). As organisational contexts become more complex and dynamic as a result of technical, economic, and demographic changes, the demand for effective training transfer increases. This bibliometrics analysis explores the nature of training transfer, the elements that drive it in today's workplaces, and the developing trends that will shape its future.

The goal of this study is to map and analyse the research landscape on transfer of training using bibliometric methodologies. The study analyses Scopus-indexed articles to assess scholarly performance in terms of annual publication output, citation counts, and author productivity. In addition, co-authorship and keyword co-occurrence analyses are used to identify and display research clusters, allowing for the examination of topic structures and collaboration patterns in the area. This method gives both a quantitative review of research performance and a qualitative analysis of the intellectual and conceptual growth of training studies. The structure of the paper is as follows. Section 1 introduces the study and outlines its background and objectives. Section 2 elaborates on the methodology and protocols used in the bibliometric analysis. Section 3 presents a descriptive analysis of the publications and citations related to the study's performance. Section 4 discusses the research clusters identified in the analysis. Section 5 explores potential future research directions, and Section 6 concludes the study.

2. Method

In the last decade, bibliometric studies have become an increasingly important tool for systematically and quantitatively examining the evolution of research fields and identifying potential areas for future investigation (Hyland & Jiang, 2021). By analysing patterns of publication output, citation trends, and collaboration networks, such studies offer insights into the intellectual structure of a discipline and its thematic development over time. Bibliometric approaches have been applied across diverse domains, including medicine, social sciences, psychology, business and management, and interdisciplinary research, helping to map scientific landscapes, identify influential contributors, and detect emerging research fronts. While a variety of analytical techniques can be applied to bibliometric data, the present study

focuses on two core metrics: the overall quantity of documents and the number of citations they have received. These measures are widely recognised as reliable indicators of research productivity and scholarly impact (Szomszor et al., 2021).

The Scopus database was selected as the primary source for the bibliometric analysis because it is widely recognised as one of the most comprehensive and authoritative platforms for indexing peer-reviewed literature. Scopus indexes over 15,000 journals and contains more than 50 million records, covering a broad range of disciplines and ensuring robust coverage of the research domain under investigation. The methodological approach adopted in this study aligns with established practices in bibliometric research and has been extensively applied in prior studies of a similar scope and nature. In addition, Scopus offers advanced search functionalities, consistent indexing standards, and high-quality metadata, which are critical for ensuring the accuracy and reliability of bibliometric analyses. Its extensive coverage also minimises the risk of omitting relevant publications, particularly in interdisciplinary areas such as transfer of training. Furthermore, the database's compatibility with bibliometric software tools, such as VOSviewer and Bibliometrix, facilitates efficient data extraction, processing, and visualisation.

A preliminary exploratory review of scholarly articles on transfer of training was conducted to assess the current state of knowledge in the field, generate a comprehensive set of relevant keywords, and compile associated terminology for subsequent analysis. The search was performed in the Scopus database using the query TITLE-ABS-KEY ("transfer of training" OR "training transfer"), which retrieves records containing the specified terms in the title, abstract, or author keywords. This search strategy ensured that the retrieved studies placed the concept at the core of their investigation rather than mentioning it only peripherally. Data was collected on 29 July 2025, producing an initial dataset of 2,754 documents. No exclusion criteria were applied, as the aim was to capture transfer of training research across all subject areas, thereby providing a comprehensive bibliometric overview that reflects the field's multidisciplinary nature and enables the identification of broader publication trends, citation patterns, and thematic developments.

3. Results

This section describes the bibliometric findings in transfer of training regarding publication and citation performances, top author contributions, productive countries, productive affiliation, productive journals, and top cited articles.

3.1 Publication and Citation of Transfer of Training Study

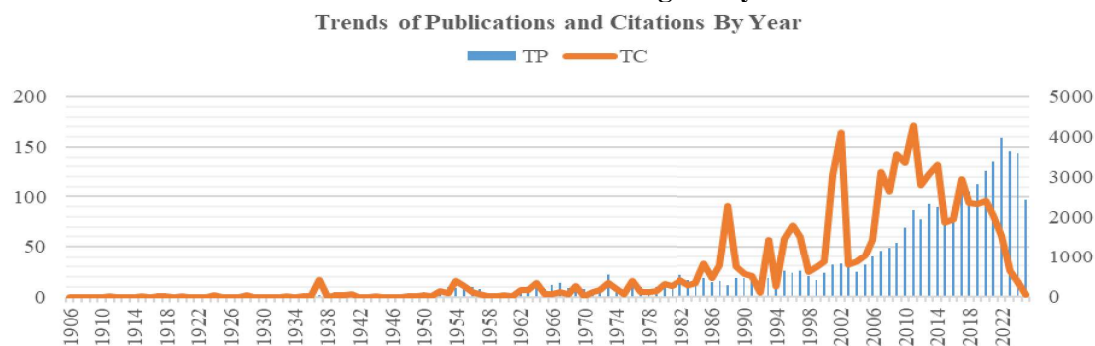


Figure 1: Articles and Citations

A descriptive statistical analysis was conducted to determine the trends of publications and citations by year. As shown in Figure 1, the number of publications remained very low from 1906 until the early 1980s, indicating minimal scholarly activity on the topic during that period. A noticeable increase in publication volume began in the late 1980s, with a sharp upward trend emerging from the mid-1990s onward. The most substantial growth occurred between 2005 and 2022, reflecting a significant rise in research interest. The peak in publication output was recorded in 2022, followed by a slight decline in subsequent years. Citations also remained low until the 1980s but began rising more significantly from the late 1980s onward. Several peaks in total citations are observable around 1991, 1999, 2008, and especially between 2010 and 2016, suggesting the presence of influential publications during these periods. While total citations increased in parallel with publications up to 2019, a marked decline is evident from 2020 onwards. This drop is likely due to the shorter time window for newly published works to accumulate citations. Overall, the field demonstrates sustained growth in scholarly output, particularly in the last two decades, accompanied by a citation pattern that reflects the time-dependent nature of academic impact.

3.2 Most Cited Papers

Citation analysis was conducted to identify the most productive and effective authors in transfer of training. Top cited articles explain the article that receives the highest citations. The top 20 cited articles performed well, with between 435 and 2358 citations. Based on Table 1, the top cited article belongs N.E. Seymour, A.G. Gallagher, S.A. Roman, M.K. O'Brien, V.K. Bansal, D.K. Andersen, R.M. Satava (2002) article on “Virtual reality training improves operating room performance results of a randomized, double-blinded study.” This article received 2358 citations, an average of 102.52 per year. The second most cited article belongs to T.T. Baldwin, J.K. Ford (1988) article on “Transfer of training: a review and directions for future research”. This article received 1972 citations, an average of 53.3 per year. The third most cited article is E. Salas, J.A. Cannon-Bowers (2001) article on “The science of training: a decade of progress.” This article received 1116 citations, an average of 46.5 per year. This is followed by B.D. Blume, J.K. Ford, T.T. Baldwin, and J.L. Huang’s “Transfer of training: a meta-analytic review”, which received 996 citations (66.4 citations annually), and H. Aguinis and K. Kraiger’s (2009) “Benefits of training and development for individuals and teams, organizations, and society”, cited 878 times with an annual average of 54.88 citations. The chronological distribution shows that while a few influential works date back to the late 1980s and early 1990s, the majority were published after 2000, pointing to a sustained period of conceptual refinement and empirical validation over the last two decades. This suggests that the field continues to evolve, with consistent citation rates indicating ongoing applicability of these findings to contemporary training practices.

Table 1: Most cited article

No.	Author(s)	Year	Title	Total Citation	Cites Per Year
1.	N.E. Seymour, A.G. Gallagher, S.A. Roman, M.K. O'Brien, V.K. Bansal, D.K. Andersen, R.M. Satava	2002	Virtual reality training improves operating room performance results of a randomized, double-blinded study	2358	102.52
2.	T.T. Baldwin, J.K. Ford	1988	Transfer of training: a review and directions for future research	1972	53.3
3.	E. Salas, J.A. Cannon-Bowers	2001	The science of training: a decade of progress	1116	46.5
4.	B.D. Blume, J.K. Ford, T.T. Baldwin, J.L. Huang	2010	Transfer of training: a meta-analytic review	996	66.4

5.	H. Aguinis, K. Kraiger	2009	Benefits of training and development for individuals and teams, organizations, and society	878	54.88
6.	L.A. Burke, H.M. Hutchins	2007	Training transfer: an integrative literature review	869	48.28
7.	Z. Shipstead, T.S. Redick, R.W. Engle	2012	Is working memory training effective?	857	65.92
8.	C. Basak, W.R. Boot, M.W. Voss, A.F. Kramer	2008	Can training in a real-time strategy video game attenuate cognitive decline in older adults?	650	38.24
9.	T. Sitzmann, K. Ely	2011	A meta-analysis of self-regulated learning in work-related training and educational attainment: what we know and where we need to go	644	46
10.	J.A. Adams	1987	Historical review and appraisal of research on the learning, retention, and transfer of human motor skills	565	14.87
11.	R. Grossman, E. Salas	2011	The transfer of training: what really matters	556	39.71
12.	E.F. Holton	1996	The flawed four-level evaluation model	538	18.55
13.	S. Moreno, C. Marques, A. Santos, M. Santos, S.L. Castro, M. Besson	2009	Musical training influences linguistic abilities in 8-year-old children: more evidence for brain plasticity	529	33.06
14.	J.B. Tracey, S.I. Tannenbaum, M.J. Kavanagh	1995	Applying trained skills on the job: the importance of the work environment	522	17.4
15.	T.S. Redick, Z. Shipstead, T.L. Harrison, K.L. Hicks, D.E. Fried, D.Z. Hambrick, M.J. Kane, R.W. Engle	2013	No evidence of intelligence improvement after working memory training: a randomized, placebo-controlled study	521	43.42
16.	H.S. Seung, H. Sompolinsky	1993	Simple models for reading neuronal population codes	481	15.03
17.	L.P. Sturm, J.A. Windsor, P.H. Cosman, P. Cregan, P.J. Hewett, G.J. Maddern	2008	A systematic review of skills transfers after surgical simulation training	479	28.18
18.	J. Karbach, P. Verhaeghen	2014	Making working memory work: a meta-analysis of executive-control and working memory training in older adults	462	42
19.	J.Z. Rouiller, I.L. Goldstein	1993	The relationship between organizational transfer climate and positive transfer of training	455	14.22
20.	K.W. Spence	1937	The differential response in animals to stimuli varying within a single dimension	435	4.94

3.3 Countries with The Highest Number of Articles

This dataset reveals a clear dominance of a few countries in research output. The United States is far ahead of all others, producing 1,032 publications over a third (35%) of the total. The United Kingdom follows with 236 publications (8%), and Germany with 167 (5.7%). This wide gap shows the overwhelming lead of the U.S. in driving research activity, with other countries contributing significantly less to comparison. The distribution of publications follows a highly unequal pattern. The top three countries alone account for nearly half of the total, while most countries in the dataset contribute under 1% each. Middle-tier contributors like Canada, Australia, and the Netherlands play important roles but still fall well below the top performers. This indicates a concentration of research capabilities in a small group of nations, likely linked to stronger funding systems, established academic infrastructure, and global research networks. Overall, the data follows a Pareto-like pattern: a small number of countries produce most of the research output, while the majority contribute only marginally. The top 10 countries alone

likely account for well over two-thirds of total publications. This concentration suggests that access to resources, research networks, and international collaborations remains unevenly distributed, reinforcing the dominance of established research powerhouses in shaping the global academic landscape.

Table 2: Countries with the most publications

No.	Country	Total Publication	Percentage (10%)
1	United States	1032	35.16184
2	United Kingdom	236	8.040886
3	China	220	7.495741
4	Germany	167	5.689949
5	Canada	165	5.621806
6	Australia	131	4.463373
7	Netherlands	105	3.577513
8	India	73	2.487223
9	Spain	65	2.214651
10	Malaysia	53	1.805792
11	Japan	50	1.703578
12	Italy	43	1.465077
13	Belgium	39	1.32879
14	Sweden	33	1.124361
15	Switzerland	33	1.124361
16	France	31	1.056218
17	Brazil	26	0.88586
18	New Zealand	25	0.851789
19	Russian Federation	25	0.851789
20	South Korea	24	0.817717
21	Ireland	22	0.749574
22	Norway	22	0.749574
23	Austria	20	0.681431
24	Taiwan	20	0.681431
25	Thailand	20	0.681431
26	Israel	19	0.647359
27	Denmark	18	0.613288
28	Finland	18	0.613288
29	Greece	18	0.613288
30	Singapore	17	0.579216
31	Portugal	15	0.511073
32	Saudi Arabia	15	0.511073
33	Hong Kong	14	0.477002
34	Indonesia	13	0.44293
35	Iran	13	0.44293
36	Pakistan	12	0.408859
37	United Arab Emirates	12	0.408859
38	Poland	11	0.374787
39	South Africa	11	0.374787
40	Mexico	10	0.340716

41	Nepal	9	0.306644
42	Uganda	9	0.306644
43	Czech Republic	7	0.238501
44	Slovenia	7	0.238501
45	Turkey	7	0.238501

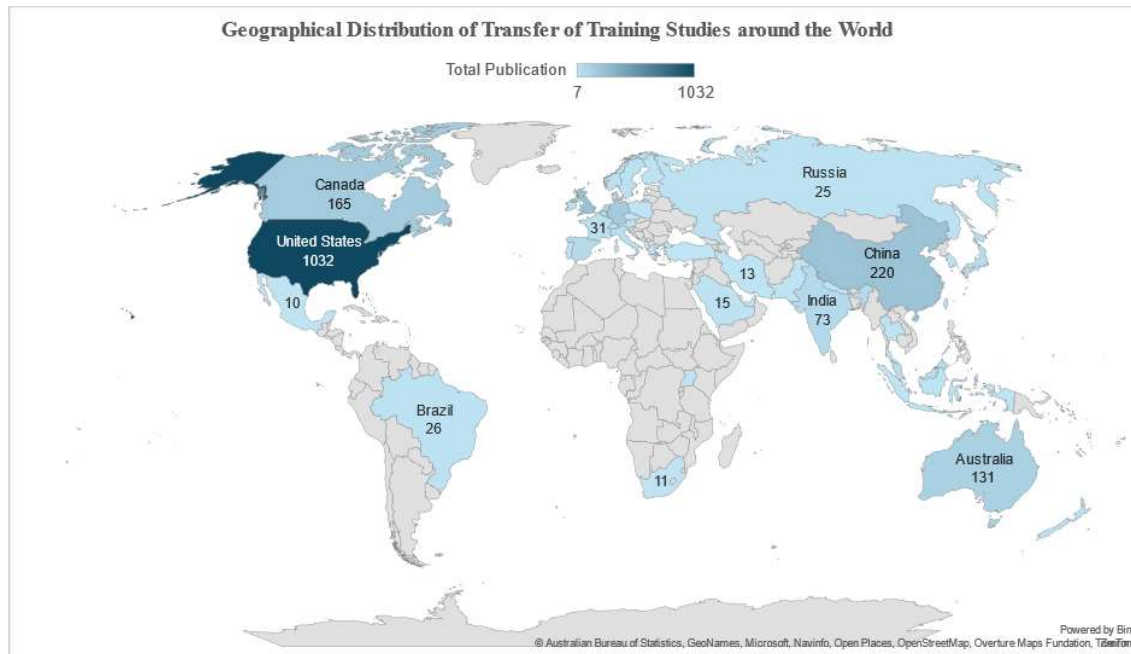


Figure 2: Geographical Distribution of Transfer of Training Studies around the World

3.4 Journals with The Highest Number of Articles and Citations

The list shows that International Journal of Training and Development has the highest number of total publications (71) among the top journals, with 62 of them cited, resulting in a strong citation count of 3,379. This translates to 47.6 citations per publication and an even higher 54.5 citations per cited publication, a sign of both productivity and consistent impact. The Journal of Experimental Psychology follows with 59 publications and a more modest citation rate of about 15 citations per publication, suggesting steady influence but not as high per-article impact as the leaders. Interestingly, Human Resource Development Quarterly stands out for quality rather than quantity. It has only 40 publications, but every one of them is cited, with a total of 3,332 citations. That's an exceptional 83.3 citations per publication, the highest in the list, indicating that its articles are widely read and referenced. Frontiers in Psychology shows a similar trend: all 21 of its publications are cited, with an impressive 63.8 citations per publication, placing it among the top in impact despite its smaller output.

Other journals, such as Human Factors and the Journal of Applied Psychology, also combine relatively moderate output with strong average citation rates (over 44 and 46 citations per publication, respectively). In contrast, Proceedings of the Human Factors and Ergonomics Society has a higher volume of work (46 publications) but a very low citation rate (3.5 per publication), showing that output alone doesn't guarantee influence. Overall, the data reflects a pattern where some journals excel in both volume and influence, while others achieve their standing through fewer but highly impactful publications.

Table 3: Journals with the highest number of articles and citations

No	Journal Title	Total publication (TP)	Number of Cited Publications (NCP)	Total Citations (TC)	Citations per Publication (TC/TP)	Citations per Cited Publication (TC/NCP)
1.	International Journal of Training and Development	71	62	3379	47.591549	54.5
2.	Journal of Experimental Psychology	59	49	887	15.033898	18.10204
3.	Proceedings of the Human Factors and Ergonomics Society	46	33	161	3.5	4.878788
4.	Human Resource Development Quarterly	40	40	3332	83.3	83.3
5.	Human Factors	33	32	1471	44.575758	45.96875
6.	Journal of Educational Psychology	30	26	281	9.3666667	10.80769
7.	European Journal of Training and Development	26	26	622	23.923077	23.92308
8.	Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics	23	19	134	5.826087	7.052632
9.	Frontiers in Psychology	21	21	1340	63.809524	63.80952
10.	Journal of Applied Psychology	20	19	937	46.85	49.31579

3.5 Affiliations with The Highest Number of Articles

The University of Illinois Urbana-Champaign leads the ranking with 50 publications, representing about 6.46% of the total output in this dataset. It is followed by the University of Toronto (35 publications, 4.52%) and the University of Central Florida (32 publications, 4.13%). Several other U.S. institutions, including Michigan State University, University of Minnesota Twin Cities, and The Ohio State University, appear prominently in the top 10, underscoring the dominance of North American universities in research output. Together, the top five institutions contribute over 20% of total publications. While U.S. and Canadian universities make up much of the upper tier, there is a significant international presence. Delft University of Technology (Netherlands), UNSW Sydney (Australia), and Universiteit Maastricht (Netherlands) rank within the top 10, each contributing between 2.45% and 2.71% of the total.

Other global research leaders in the list include KU Leuven (Belgium), Technische Universität München (Germany), and The University of Tokyo (Japan), showing that impactful research contributions are not limited to North America. The lower half of the list shows a clustering of institutions with 1.3%–2% contribution each, including well-known names such as King's College London, Karolinska Institutet, and Johns Hopkins University. This distribution reflects a long tail pattern in which a few institutions dominate in output, while many others contribute modest but steady numbers. The spread of institutions across multiple continents suggests a broad international research network, even though the publication output is still heavily weighted toward a small set of highly productive universities.

Table 4: Affiliations with the highest number of publications

No.	Affiliation	Total Publication (TP)	Percentage (10%)
1.	University of Illinois Urbana-Champaign	50	6.459948
2.	University of Toronto	35	4.521964
3.	University of Central Florida	32	4.134367
4.	Michigan State University	29	3.74677
5.	University of Minnesota Twin Cities	23	2.971576
6.	The Ohio State University	21	2.713178
7.	Delft University of Technology	21	2.713178
8.	UNSW Sydney	20	2.583979
9.	Universiteit Maastricht	19	2.45478
10.	University of Michigan, Ann Arbor	19	2.45478
11.	Indiana University Bloomington	17	2.196382
12.	Universitat Autònoma de Barcelona	17	2.196382
13.	University of Wisconsin-Madison	17	2.196382
14.	University of Houston	16	2.067183
15.	University of Pittsburgh	16	2.067183
16.	The University of Sheffield	15	1.937984
17.	Florida State University	15	1.937984
18.	Universiteit van Amsterdam	15	1.937984
19.	Louisiana State University	15	1.937984
20.	University of Michigan-Flint	15	1.937984
21.	Pennsylvania State University	14	1.808786
22.	Old Dominion University	14	1.808786
23.	Texas A&M University	14	1.808786
24.	KU Leuven	14	1.808786
25.	Technische Universität Braunschweig	13	1.679587
26.	Purdue University	13	1.679587
27.	Technische Universität München	13	1.679587
28.	The University of Queensland	13	1.679587
29.	McMaster University	13	1.679587
30.	University of South Florida, Tampa	12	1.550388
31.	University of Washington	12	1.550388
32.	The University of Tennessee, Knoxville	12	1.550388
33.	Memorial University of Newfoundland	12	1.550388
34.	University of Colorado Boulder	11	1.421189
35.	Ministry of Education of the People's Republic of China	11	1.421189
36.	The University of Manchester	11	1.421189
37.	Johns Hopkins University	11	1.421189
38.	Colorado State University	11	1.421189
39.	The University of Tokyo	11	1.421189
40.	Embry-Riddle Aeronautical University	11	1.421189
41.	University at Buffalo, The State University of New York	11	1.421189
42.	Universiti Teknologi MARA	10	1.29199
43.	Turun yliopisto	10	1.29199

44.	King's College London	10	1.29199
45.	Karolinska Institutet	10	1.29199
46.	San Jose State University	10	1.29199
47.	University of Kentucky	10	1.29199
48.	Georgia Institute of Technology	10	1.29199
49.	Netherlands Organisation for Applied Scientific Research - TNO	10	1.29199
50.	University of Liverpool	10	1.29199

3.6 Authors with The Highest Number of Papers

The author dataset contains 65 contributors, with publication counts ranging from 5 to 20. The most prolific is A. Gegenfurtner, producing 20 publications (4.30%), followed by B.D. Blume and S. Kauffeld, each with 14 publications (3.01%). G. Lintern ranks next with 12 publications (2.58%), while four authors namely W.R. Boot, K. Nielsen, C. Quesada-Pallarès, and E. Salas, each have 11 publications (2.37%). A mid-tier group of eight authors, including H.M. Hutchins (10 publications) and R. Bates (9 publications), contribute between 9 and 10 publications each, representing 1.94–2.15% individually. This group plays a substantial role in shaping the research output, even though they publish less frequently than the leading authors. The largest share of contributors falls within the lower publication bracket of 5–8 papers, with 42 out of 65 authors in this range. This long-tail distribution reflects a common trend in academic publishing, where a small number of highly productive authors account for a significant portion of total output, while the majority contribute more modestly.

Table 5: Authors with the highest number of articles

No.	Author Name	Total Publication (TP)	Percentage (10%)
1.	Gegenfurtner, A.	20	4.301075
2.	Blume, B.D.	14	3.010753
3.	Kauffeld, S.	14	3.010753
4.	Lintern, G.	12	2.580645
5.	Boot, W.R.	11	2.365591
6.	Nielsen, K.	11	2.365591
7.	Quesada-Pallarès, C.	11	2.365591
8.	Salas, E.	11	2.365591
9.	Hutchins, H.M.	10	2.150538
10.	Bates, R.	9	1.935484
11.	Burke, L.A.	9	1.935484
12.	Chiaburu, D.S.	9	1.935484
13.	Huang, J.L.	9	1.935484
14.	Kramer, A.F.	9	1.935484
15.	Mulder, M.	9	1.935484
16.	Saks, A.M.	9	1.935484
17.	Gagné, R.M.	8	1.72043
18.	Pool, D.M.	8	1.72043
19.	Besson, M.	7	1.505376
20.	Bongers, R.M.	7	1.505376
21.	Brown, T.C.	7	1.505376
22.	Ettlinger, G.	7	1.505376
23.	Ford, J.K.	7	1.505376
24.	Hancock, P.A.	7	1.505376

25.	Karbach, J.	7	1.505376
26.	McLean, G.N.	7	1.505376
27.	Zumrah, A.R.	7	1.505376
28.	de Jong, B.	7	1.505376
29.	Baldwin, T.T.	6	1.290323
30.	Boninger, M.L.	6	1.290323
31.	Cornelissen, F.	6	1.290323
32.	Dux, P.E.	6	1.290323
33.	Edwards, J.D.	6	1.290323
34.	Elliott, D.	6	1.290323
35.	Holton, E.	6	1.290323
36.	Ismail, A.	6	1.290323
37.	Jacobsson, L.	6	1.290323
38.	Jansen in de Wal, J.	6	1.290323
39.	Kehoe, E.J.	6	1.290323
40.	Lauzier, M.	6	1.290323
41.	Mania, K.	6	1.290323
42.	Park, S.	6	1.290323
43.	Pineda-Herrero, P.	6	1.290323
44.	Vignoli, M.	6	1.290323
45.	Voss, M.W.	6	1.290323
46.	Arasanmi, C.N.	5	1.075269
47.	Baker, K.E.	5	1.075269
48.	Baltes, P.B.	5	1.075269
49.	Bhatti, M.A.	5	1.075269
50.	Dixit, R.	5	1.075269
51.	Eggert, F.	5	1.075269
52.	Foster, H.	5	1.075269
53.	Ghosh, P.	5	1.075269
54.	Gruber, H.	5	1.075269
55.	Hasson, H.	5	1.075269
56.	Kanai-Pak, M.	5	1.075269
57.	Kevin Ford, J.	5	1.075269
58.	Kitajima, Y.	5	1.075269
59.	Kuwahara, N.	5	1.075269
60.	Lim, D.H.	5	1.075269
61.	Maeda, J.	5	1.075269
62.	Miuro, R.F.	5	1.075269
63.	Mouloua, M.	5	1.075269
64.	Müller-Ruchholtz, W.	5	1.075269
65.	Nakamura, M.	5	1.075269

4. Research Stream / Cluster / Trend

The keyword co-occurrence analysis using VOSviewer produced six distinct clusters, each representing a coherent thematic area within the transfer of training literature. These clusters were identified through the software's mapping and clustering algorithms, which group terms

based on their frequency of co-occurrence and link strength. The themes were assigned by examining the most frequent and central keywords in each cluster, supported by relevant theoretical and empirical literature to ensure conceptual validity. Table X presents the derived themes, the dominant keywords, and their interpretations. Collectively, these themes reflect both traditional perspectives on training transfer such as motivational and environmental determinants—and emerging areas of research, including immersive technologies, cognitive mechanisms, and meta-analytic syntheses. Table 6. In addition, Figure 2 shows the illustration of keyword network analysis using VosViewer software.

Table 6: Cluster's Analysis Matrix

Cluster No.	Items (Labels) and Occurrences	No. of Items	Total Link Strengths	Total Occurrence	Interpretation
1. Determinants of Training Transfer	motivation to learn (17), motivation to transfer (33), self-efficacy (30), supervisor support (32), training design (17), training effectiveness (32), training evaluation (38), training transfer (222), transfer motivation (17), transfer of learning (20), transfer of training (332), work environment (18)	12	716	808	This cluster represents a complete framework of the major characteristics empirically connected to training transfer, notably in organisational and workplace learning environments. The items are consistent with existing models such as Baldwin and Ford (1988), Holton's Transfer of Training Model (1996), and the Learning Transfer System Inventory (LTSI).
2. AI and Immersive Technologies for Training and Skill Transfer	convolutional neural network (18), deep learning (50), machine learning (22), rehabilitation (37), stroke (21), transfer learning (85), transfer training (16), virtual reality (44)	8	203	293	This cluster reflects the emerging role of artificial intelligence and immersive technologies (e.g., virtual reality, machine learning, deep learning) in enhancing the transfer of training and skills across contexts
3. Cognitive Function and Training Interventions in Learning Transfer	aging (19), attention (17), cognition (16), cognitive training (54), working memory (27)	5	172	133	This cluster centers on the role of cognitive capabilities—such as attention, working memory, and general cognition—as critical individual characteristics influencing the effectiveness and transfer of training. It also highlights the importance of cognitive training interventions, particularly in contexts involving aging populations, where maintaining or enhancing learning capacity is essential
4. Cognitive Mechanisms Underpinning Learning and Transfer	learning (61), learning & memory (17), memory (28), transfer (83)	4	192	189	This cluster represents the core theoretical and psychological constructs that underpin how knowledge and skills are acquired, retained, and transferred across tasks or contexts. It connects cognitive science and educational psychology perspectives that form the basis for understanding why and how transfer occurs

5. Applied Learning and Simulation in Training Transfer	knowledge transfer (18), simulation (28), training (206)	3	212	252	This cluster centers on practical mechanisms and environments that facilitate the application of training in real-world settings, particularly via simulations and structured learning environments designed to replicate job conditions. It links instructional design theory, experiential learning, and organizational learning transfer.
6. Meta-Analytic Perspectives on Motivation and Transfer	learning transfer (21), meta-analysis (18), motivation (19)	3	77	58	This cluster appears to synthesize the empirical literature examining how motivation influences learning transfer, drawing heavily on meta-analytic studies. It suggests a theoretical and quantitative approach to understanding what drives effective application of learning in workplace or training contexts

4.1 Determinants of Training Transfer

This cluster encompasses the key factors that shape the successful application of training, including motivation to learn, self-efficacy, supervisor support, training design, training effectiveness, and work environment. The high frequency of transfer of training (332) and training transfer (222), combined with the highest total link strength (716) and occurrence count (808), underscores the centrality of this theme in workplace learning research. These factors align with established frameworks such as Baldwin and Ford's Transfer of Training Model (Baldwin & Ford, 1988) and Holton's (1996) HRD Evaluation and Research Model (Holton III, 1996), both of which emphasise the dynamic interplay between trainee characteristics, program design, and organisational context. The co-occurrence analysis demonstrates strong relationships between motivation to learn (Kodwani & Prashar, 2019) and training design (Gegenfurtner et al., 2020), implying that learners' attitudes and the quality of instructional preparation both influence information retention and skill application. Furthermore, the frequent association of supervisor support with the work environment demonstrates the importance of organisational climate in reinforcing training outcomes (Maharmeh, 2021). This wide range of interconnected characteristics explains why this cluster serves as the conceptual underpinning of training transfer research, since effective transfer is rarely due to a single determinant but rather results from a systemic alignment of human, instructional, and organisational variables. This cluster's continuing dominance in terms of occurrence and link strength demonstrates its long-term relevance across sectors and contexts.

4.2 AI and Immersive Technologies for Training and Skill Transfer

This cluster reflects the growing integration of emerging technologies virtual reality, deep learning, machine learning, and rehabilitation in facilitating training transfer. The prominence of transfer learning (85) and virtual reality (44), supported by a total link strength of 203 and occurrence count of 293, points to an increasing research focus on technology-mediated skill acquisition across diverse domains. The convergence of artificial intelligence and immersive environments represents an innovation-driven approach to training design, supported by empirical evidence that virtual simulations can improve skill retention and transfer by providing realistic, interactive learning environments (Khabarov & Volegzhanina, 2018). The co-occurrence of AI-related and simulation-related terms indicates that researchers are not only exploring technological applications in isolation but are actively investigating their synergistic

potential for enhancing learning outcomes (Cooper et al., 2021). Notably, virtual reality is commonly used alongside terms like rehabilitation and skill retention, implying cross-sector uses in healthcare, manufacturing, and safety-critical industries (Kaizer et al., 2020; Wanasinghe et al., 2021). The incorporation of immersive technology into training frameworks also facilitates adaptive learning, which allows instruction to be tailored in real time depending on learner performance data. This pattern indicates a path towards hybridised, technology-enhanced learning ecosystems that can change in response to industry needs and AI advances.

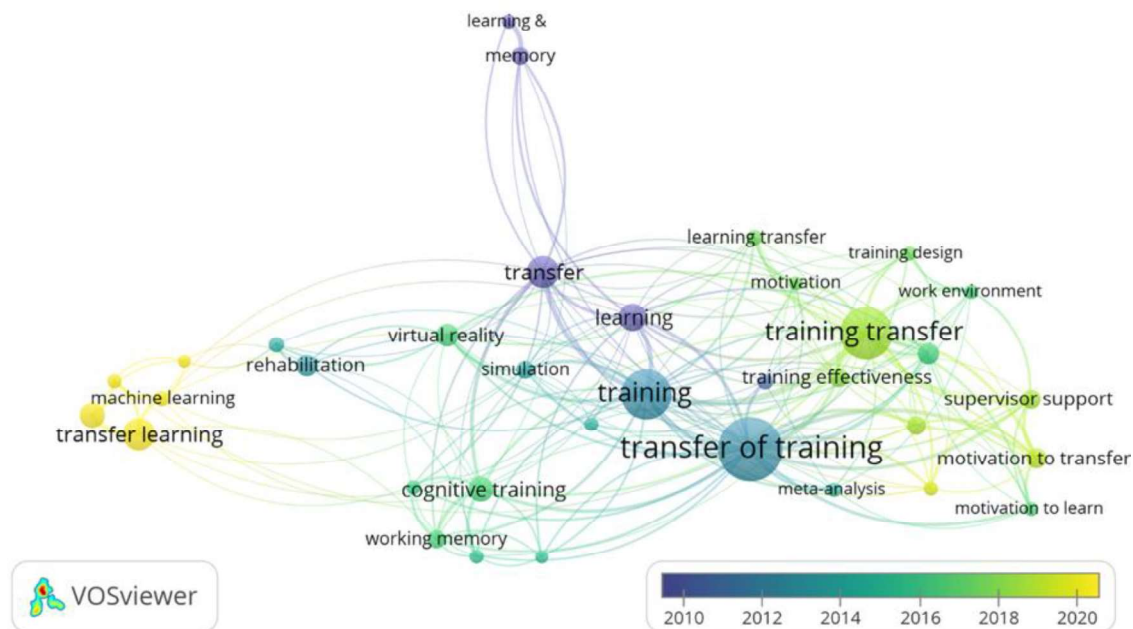


Figure 3: Keyword network analysis on Transfer of Training studies (Visualization by authors using VosViewer Software)

4.3 Cognitive Function and Training Interventions in Learning Transfer

This cluster highlights the influence of cognitive capacities such as attention, working memory, and general cognition on training transfer. With cognitive training (54) and working memory (27) as central terms, and a total link strength of 172 and occurrence count of 133, it underscores the importance of targeted interventions to enhance cognitive readiness, particularly among aging workforces. Evidence from workplace training and cognitive neuroscience research indicates that cognitive enhancement strategies can sustain or even improve learning capacity, which in turn facilitates skill retention and application in real-world tasks (Scorgie et al., 2024). Co-occurrence patterns reveal strong associations between working memory and learning retention, indicating that cognitive capacity acts as a mediating variable in the transfer process. Furthermore, the link between attention and training design underlines the need for instructional methods that minimise cognitive overload and optimise focus (Bhatti et al., 2021). This cluster is also consistent with the growing interest in customised learning paths, in which training is tailored to learners' cognitive profiles to enhance transfer potential. These findings suggest that businesses looking to improve training ROI should incorporate cognitive evaluation and enhancement tactics into their overall learning and development initiatives.

4.4 Cognitive Mechanisms Underpinning Learning and Transfer

This theme represents the psychological and theoretical foundations of how knowledge and skills are acquired, retained, and transferred. With learning domain (61), learning & memory (45), and transfer (83) as key terms, supported by a total link strength of 192 and occurrence count of 189, it bridges cognitive science and educational psychology perspectives. The inclusion of collaborative constructs like cognitive teaming indicates that learning transfer is increasingly viewed not only as an individual cognitive process but also as one embedded in team-based and social learning contexts (Islam, 2019). The co-occurrence analysis highlights that these mechanisms are frequently linked to problem-solving and conceptual change, suggesting that transfer effectiveness is enhanced when learners actively restructure knowledge rather than merely replicate skills (Gegenfurtner et al., 2020). This simultaneous focus on individual cognitive architecture and collective learning dynamics provides a thorough understanding of how training outcomes are internalised and used in complicated, real-world circumstances. The existence of such cognitive constructs also indicates a growing interest in multidisciplinary approaches that combine workplace learning research with insights from neuroscience (Parasuraman & McKinley, 2014) and human factors engineering (Samaka & Ally, 2015). Organisations may develop training interventions that optimise both individual and group performance by identifying the relationship between mental models, team cognition, and collaborative problem-solving.

4.5 Applied Learning and Simulation in Training Transfer

This cluster focuses on applied mechanisms and environments particularly simulation training (206) that facilitate the transfer of training to workplace contexts. With a total link strength of 212 and occurrence count of 252, it underlines the effectiveness of structured learning environments that closely replicate real job conditions, enabling experiential learning and skill application. This theme is firmly grounded in experiential learning theory (Kolb, 1984) and organisational learning strategies that aim to bridge the gap between formal instruction and workplace performance (Hughes et al., 2018). Co-occurrence mapping demonstrates several linkages between simulation and performance improvement, implying that practical, scenario-based training has measurable results (Cates et al., 2016). Furthermore, simulation is frequently seen in close proximity to high-risk industries such as aviation, oil and gas, and healthcare, where skill transfer is crucial for safety and operational excellence (Tan & Tanner, 2025). The immersive, feedback-rich aspect of simulation training also allows for repeated practice, which is essential for solidifying skills and boosting transfer fidelity. These characteristics make simulation-based learning an essential component of contemporary workforce training programs.

4.6 Meta-Analytic Perspectives on Motivation and Transfer

This cluster synthesises meta-analytic evidence on the role of motivation in training transfer. Although smaller in size, with a total link strength of 77 and occurrence count of 58, it offers a highly concentrated evidence base that consolidates findings from multiple studies to generate theory-driven insights. Research in this area consistently supports the conclusion that learner motivation significantly moderates the relationship between training inputs and post-training performance (Blume et al., 2019). The co-occurrence patterns, though narrower than in larger clusters, connect motivation with goal orientation and self-determination, highlighting the psychological drivers that encourage persistence and application of learned skills. This synthesis not only affirms established motivational theories such as self-determination theory (Kumari et al., 2021) but also provides actionable guidance for designing motivational interventions within training programs. This cluster, which distils cross-study findings, serves

as a strategic reference point for businesses looking to incorporate motivation-enhancing features into both formal training and workplace learning cultures.

5. Research Question / Future Research Direction

The study on transfer of training identified several significant areas for further research directions. These questions, organised into theoretical, practical, and methodological approaches, provide a roadmap for future research. From a theoretical perspective, it is important to investigate how cognitive processes such as memory retrieval, attention, and emotional engagement interact to influence the successful application of learned skills (Ludwikowska, 2021). Advancing transfer models by integrating motivational constructs (Saleh Moumine Abdi, 2019), cultural factors (Rawashdeh & Tamimi, 2020), and the influence of emerging work arrangements (Shaheen & Soomro, 2022) can deepen understanding of the mechanisms that drive skill retention and application. Future research should also explore how new technologies such as artificial intelligence, virtual reality or augmented reality, and machine learning that shape transfer outcomes across both technical and interpersonal skills (Rad et al., 2022; Scorgie et al., 2024).

From a practical perspective, future studies should identify the most influential training design features, such as simulation fidelity, scenario complexity, and feedback mechanisms, that predict effective transfer in real-world contexts (Saville & Foster, 2021). The application of AI-driven analytics to create personalized learning pathways tailored to cognitive profiles and motivational states offers a promising avenue. Research should also focus on adapting immersive and simulation-based methods for dynamic, high-stakes industries and developing structured reinforcement strategies such as supervisor follow-up and peer learning networks to sustain post-training application over time.

From a methodological perspective, longitudinal research tracking learners from pre-training through long-term workplace performance is needed to understand sustained transfer effects (Seeg et al., 2021). Meta-analyses segmented by industry, cultural context, and delivery mode can help clarify the most consistent predictors of transfer. Experimental comparisons between high-tech immersive methods and low-tech experiential learning approaches can assess relative cost-effectiveness and impact. Mixed-methods studies that combine quantitative performance data with qualitative insights from learners and supervisors can provide a richer understanding of the complex factors influencing training transfer.

Table 7: Future Research Directions

Perspective	No	Future Research Question	Sources
Theoretical	1.	How do cognitive processes such as memory retrieval, attention, and emotional engagement interact to influence the success of training transfer?	(Dahal & Manandhar, 2024; Humida et al., 2021)
	2.	In what ways do motivational constructs and cultural factors shape learners' ability to retain and apply skills over time?	(Pilbeam & Karanikas, 2023)
	3.	How do emerging technologies like AI, VR/AR, and machine learning alter the mechanisms underlying skill retention and application?	(Scorgie et al., 2024)
Practical	4.	Which simulation design features such as fidelity, scenario complexity, and feedback type most strongly predict transfer effectiveness in real-world settings?	(Cooper et al., 2021)
	5.	How can AI-driven analytics be used to personalize learning pathways and enhance transfer for diverse learner profiles?	(Lytovchenko et al., 2022; Markwell et al., 2023)
	6.	What strategies can organizations implement to embed post-training reinforcement into daily workflows to sustain long-term transfer?	(Çimen, 2021)
Methodological	7.	What insights can be gained from longitudinal studies tracking learners from pre-training to long-term workplace performance?	(Deckers et al., 2022)
	8.	How do transfer outcomes vary when training is delivered through high-tech immersive methods versus low-tech experiential approaches?	(Cooper et al., 2021)
	9.	What additional understanding can mixed methods designs provide by combining quantitative performance metrics with qualitative learner feedback?	(Putri & Handayani, 2021; Rahman, 2021)

6. Conclusion

This study offers a structured overview of future directions in the field of transfer of training, organized into theoretical, practical, and methodological approaches. The study has three sub-objectives. The first objective is to bibliometrically analyze the performance in terms of citations and publications of transfer of training studies. The second objective is to analyze the research themes of transfer of training studies. The third objective is to provide several recommendations for future research direction. The rationale for conducting this study derives from the deficiencies identified in previous studies on transfer of training. This study mainly uses bibliometric analysis, whereby 2754 documents from 1909 to 2025 were analyzed through descriptive analysis and keyword network analysis.

Selective findings indicated that, in terms of citations and publications, there was a noticeable increase in publication volume began in the late 1980s, with a sharp upward trend emerging from the mid-1990s onward. The most substantial growth occurred between 2005 and 2022, reflecting a significant rise in research interest. Based on the publication and citation analysis, the top cited article belongs N.E. Seymour, A.G. Gallagher, S.A. Roman, M.K. O'Brien, V.K. Bansal, D.K. Andersen, R.M. Satava (2002) article on "Virtual reality training improves operating room performance results of a randomized, double-blinded study." This article received 2358 citations, an average of 102.52 per year. In terms of countries with the most articles, The United States is far ahead of all others, producing 1,032 publications over a third (35%) of the total. The United Kingdom follows with 236 publications (8%), and Germany

with 167 (5.7%). This wide gap shows the overwhelming lead of the U.S. in driving research activity, with other countries contributing significantly less to comparison.

Furthermore, International Journal of Training and Development has the highest number of total publications (71) among the top journals, with 62 of them cited, resulting in a strong citation count of 3,379. This translates to 47.6 citations per publication and an even higher 54.5 citations per cited publication, a sign of both productivity and consistent impact. The University of Illinois Urbana-Champaign leads the ranking with 50 publications, representing about 6.46% of the total output in this dataset. The most prolific is A. Gegenfurtner, producing 20 publications (4.30%), followed by B.D. Blume and S. Kauffeld, each with 14 publications (3.01%). G. Lintern ranks next with 12 publications (2.58%), while four authors namely W.R. Boot, K. Nielsen, C. Quesada-Pallarès, and E. Salas, each have 11 publications (2.37%). The authors also proposed 9 future research questions, categorized into three distinct areas – theoretical, practical, and methodological. These research questions provide significant contributions by assisting and motivating scholars to undertake future investigations into transfer of training.

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Conflict of Interest Statement

The authors declare that there is no conflict of interest regarding the publication of this study.

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